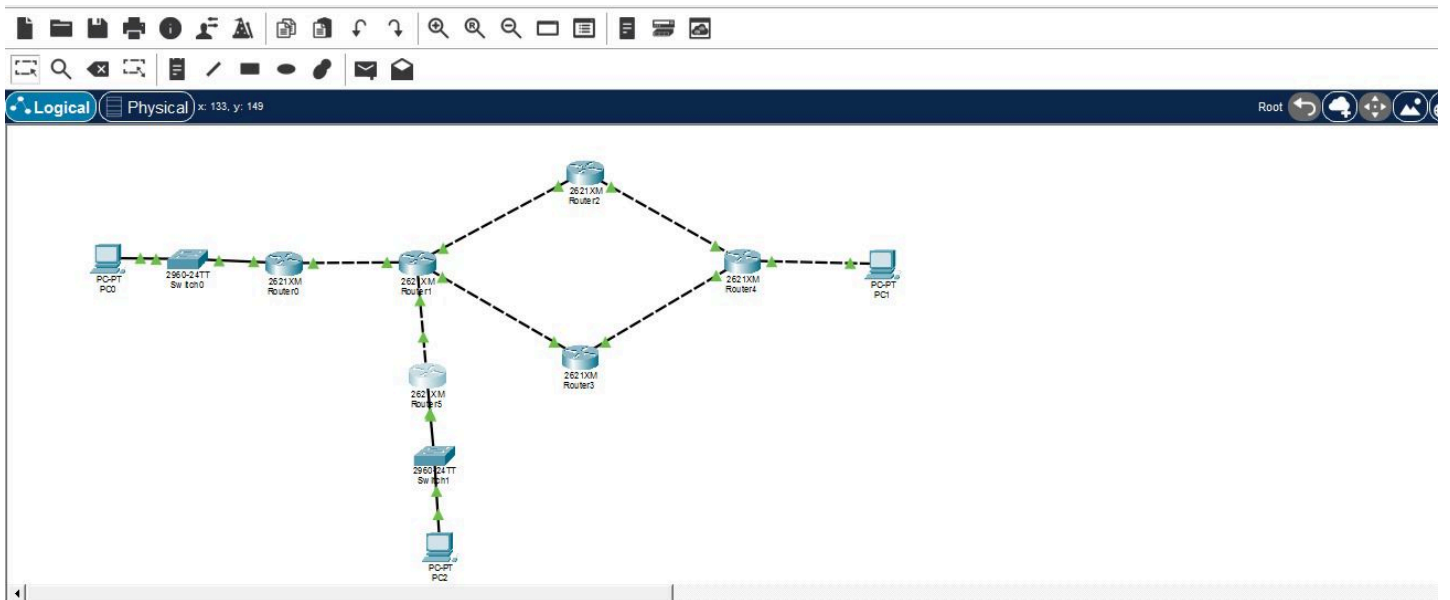




```

Router(config)#ip route 223.0.50.0 255.255.255.0 12.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 12.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    12.0.0.0/8 is directly connected, FastEthernet0/1
S    23.0.0.0/8 [1/0] via 12.0.0.2
S    24.0.0.0/8 [1/0] via 12.0.0.2
S    26.0.0.0/8 [1/0] via 12.0.0.2
S    32.0.0.0/8 [1/0] via 12.0.0.2
S    45.0.0.0/8 [1/0] via 12.0.0.2
S    172.36.0.0/16 [1/0] via 12.0.0.2
C    192.50.20.0/24 is directly connected, FastEthernet0/0
S    223.0.50.0/24 [1/0] via 12.0.0.2

Router#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.50.20.1     YES manual up         up
FastEthernet0/1 12.0.0.1        YES manual up         up
Router#

```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#ip route 32.0.0.0 255.0.0.0 24.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 24.0.0.2
Router(config)#ip route 45.0.0.0 255.0.0.0 24.0.0.2
Router(config)#ip route 45.0.0.0 255.0.0.0 23.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 24.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    12.0.0.0/8 is directly connected, FastEthernet0/0
C    23.0.0.0/8 is directly connected, FastEthernet1/0
C    24.0.0.0/8 is directly connected, FastEthernet1/1
C    26.0.0.0/8 is directly connected, FastEthernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
      [1/0] via 24.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
      [1/0] via 23.0.0.2
S    172.36.0.0/16 [1/0] via 23.0.0.2
      [1/0] via 24.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2

Router#
Router#
```

```
Router(config)#ip route 24.0.0.0 255.0.0.0 32.0.0.2
Router(config)#ip route 45.0.0.0 255.0.0.0 23.0.0.1
Router(config)#ip route 45.0.0.0 255.0.0.0 32.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 32.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.1
Router(config)#end
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#sh ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
S    12.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
C    23.0.0.0/8 is directly connected, FastEthernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
S    26.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
C    32.0.0.0/8 is directly connected, FastEthernet0/1
S    45.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
S    172.36.0.0/16 [1/0] via 32.0.0.2
      [1/0] via 23.0.0.1
S    192.50.20.0/24 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
S    223.0.50.0/24 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
```

```
Router#
```

```
Router#
```

```
S    223.0.50.0/24 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2

Router#
Router#conf t
Enter configuration commands, one per line. End with CNIL/Z.
Router(config)#ip route 26.0.0.0 255.0.0.0 24.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2
S    23.0.0.0/8 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2
C    24.0.0.0/8 is directly connected, FastEthernet0/0
S    26.0.0.0/8 [1/0] via 45.0.0.2
      [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2
C    45.0.0.0/8 is directly connected, FastEthernet0/1
S    172.36.0.0/16 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2
S    223.0.50.0/24 [1/0] via 24.0.0.1
      [1/0] via 45.0.0.2
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.36.0.10, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#
Router#
Router#
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
C    32.0.0.0/8 is directly connected, FastEthernet0/1
C    45.0.0.0/8 is directly connected, FastEthernet1/0
C    172.36.0.0/16 is directly connected, FastEthernet0/0
S    192.50.20.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1

Router#
Router#
Router#
Router#
Router#
```

```

FastEthernet0/1      26.0.0.2      YES manual up      up
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.50.20.0 255.255.255.0 26.0.0.1
Router(config)#ip route 12.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 23.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 24.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 32.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 45.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 172.36.0.0 255.255.0.0 26.0.0.1
^
% Invalid input detected at '^' marker.

Router(config)#ip route 172.36.0.0 255.255.0.0 26.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
C    26.0.0.0/8 is directly connected, FastEthernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
C    223.0.0.0/8 is directly connected, FastEthernet0/0

Router#

```

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=10ms TTL=124
Reply from 172.36.0.10: bytes=32 time=10ms TTL=120
Reply from 172.36.0.10: bytes=32 time=10ms TTL=124
Reply from 172.36.0.10: bytes=32 time=10ms TTL=120

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 10ms, Average = 10ms

C:\>
C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

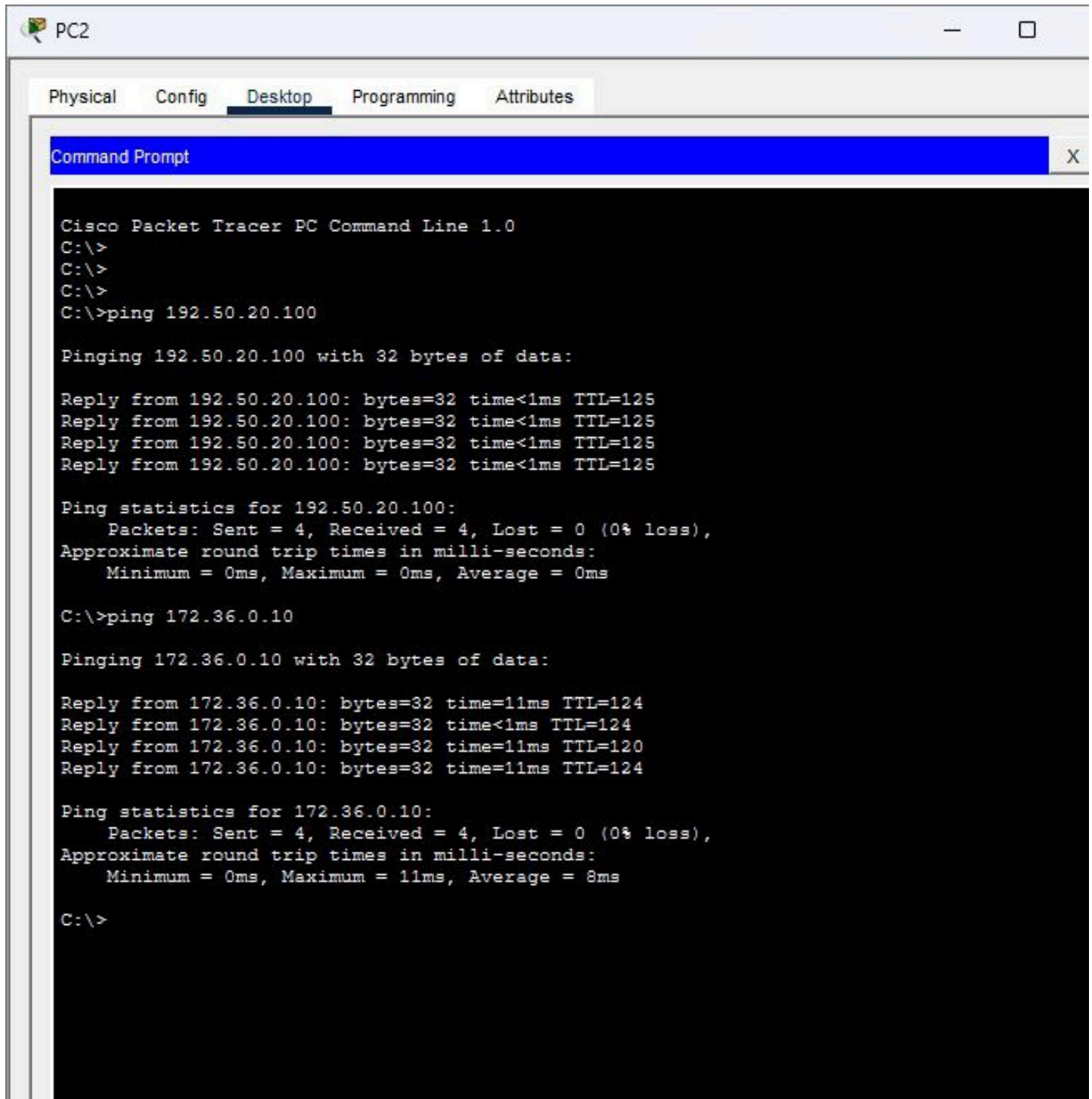
C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=19ms TTL=124
Reply from 172.36.0.10: bytes=32 time=11ms TTL=120
Reply from 172.36.0.10: bytes=32 time=10ms TTL=124
Reply from 172.36.0.10: bytes=32 time=14ms TTL=120

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 19ms, Average = 13ms

C:\>
```



The screenshot shows a window titled "PC2" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of two ping commands: one to 192.50.20.100 and another to 172.36.0.10. Both pings are successful with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>
C:\>
C:\>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=11ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time=11ms TTL=120
Reply from 172.36.0.10: bytes=32 time=11ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 8ms

C:\>
```

Command Prompt

```
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=11ms TTL=120
Reply from 223.0.50.10: bytes=32 time=10ms TTL=124
Reply from 223.0.50.10: bytes=32 time=10ms TTL=120
Reply from 223.0.50.10: bytes=32 time=11ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 11ms, Average = 10ms

C:\>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=10ms TTL=120
Reply from 192.50.20.100: bytes=32 time=10ms TTL=124
Reply from 192.50.20.100: bytes=32 time=10ms TTL=120
Reply from 192.50.20.100: bytes=32 time=12ms TTL=124

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 12ms, Average = 10ms

C:\>
```